Sahaj Bhadja

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Summary

Systems-focused software engineer with 2+ years building and debugging performance-critical software using C++, Python, multithreading, and distributed services. Strong foundation in operating systems, algorithms, runtime analysis, automated validation, cross-platform reliability, and AI-assisted engineering.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built production Python services for 500+ users, profiling execution paths and optimizing indexed queries, schemas, and caching to reduce response latency from 1.2 seconds to 350 milliseconds.
- Investigated failures in a concurrent job orchestration system, introducing explicit runtime states, failure isolation, diagnostics, and recovery controls that tripled throughput while improving repeatability and production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Python services handling 10,000+ daily requests, tracing production failures through logs, source code, state transitions, and error paths while resolving reliability issues affecting more than 150 users.
- Engineered multithreaded processing for 1,000+ files with bounded workers, automated validation, progress instrumentation, and failure recovery, reducing turnaround 60% while maintaining stable behavior under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python and PyTorch analysis pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, runtime profiling, automated validation, model evaluation, and systematic root-cause debugging.
- Reconstructed unstable simulation behavior from partial logs and agent state, improving closed-loop stability 15% while documenting evidence, assumptions, experiments, confidence levels, and reproducible technical findings for researchers.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Profiled Python services and data-processing integrations using AWS Glue, S3, SQL, pagination, and caching, improving production response times 30% and processing efficiency 68% across connected enterprise applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating defect isolation, regression verification, and reliable releases across Windows, Linux, and macOS production environments.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular services across eight microservices with authentication, WebSocket messaging, state synchronization, and defensive error handling, securely processing more than two million records through documented software interfaces.
- Automated builds, functional tests, staged deployments, and production diagnostics through Jenkins, Docker, and Azure, reducing release time 50% while improving validation coverage, deployment consistency, and operational visibility.

Skills

Languages & Fundamentals: C++, Python, Java, SQL, data structures, algorithms, object-oriented programming, computer architecture fundamentals, operating systems concepts
Systems & Runtime: Multithreading, concurrency, distributed services, Linux, Windows, macOS, process and thread concepts, memory management fundamentals, runtime profiling, performance analysis
Diagnostics & Quality: Source-level debugging, log analysis, root-cause analysis, automated validation, unit testing, integration testing, PyTest, Mockito, monitoring, failure isolation, regression testing
Engineering & Automation: Git, Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, scripting, technical documentation, cross-functional collaboration, production support, AI-assisted workflows

Projects

CPU Scheduling Simulator | Live | *C++, React, Operating Systems, Algorithms*
- Implemented six CPU scheduling algorithms in C++ with interactive controls and Gantt visualization, validating edge cases and comparing execution, waiting, and turnaround metrics across 1,000+ process combinations.

Chat2SQL | Code | *Java, TypeScript, SQL, Docker, Maven*
- Built a modular database tool supporting 10+ engines, reasoning across unfamiliar schemas and error traces while adding query validation, metadata inspection, automated tests, and Docker-backed reproducible deployment.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a client-server platform supporting 50,000+ users and one million transactions, integrating secure REST APIs, WebSockets, MongoDB persistence, Kubernetes deployment, production monitoring, and automated release workflows.